

What changes the S4 / MLP comparison?

Smooth source, unstructured head

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The original experiment did not show the proposed GP-source MLP disadvantage. This follow-up tests two possible reasons: the weighting of small-variance directions during training, and the limited information supplied by a roughness score. It adds a controlled head-by-loss experiment and a fresh conditional-forecast evaluation.

New evidence	Completed design
Training control	MLP / S4 × raw / balanced loss; identical full normalized history; 3 final seeds per cell
Validation search	8 pilots, then locked learning rates before final test evaluation
Forecast evaluation	15 original GP checkpoints; 256 fresh histories × 64 paths; Euler-128
Numerical checks	Independent S4 kernel, CPU/MPS parity, saved-array recomputation, Euler-256 sensitivity

The comparison depends on the objective. Normalized residual MSE favors MLP under raw loss (0.222 versus 31.122), but S4 under balanced loss (0.158 versus 0.175). S4 also preserves PC1 better than the balanced MLP. Raw residual MSE still favors MLP under both losses.

A tradeoff, not a universal winner. Balancing changes S4’s generated/target small-direction energy ratio from 597.2 to 0.585, removing the excess but now undershooting the target. Its mean raw PC1 and residual errors increase. Good roughness alone did not reveal this behavior.

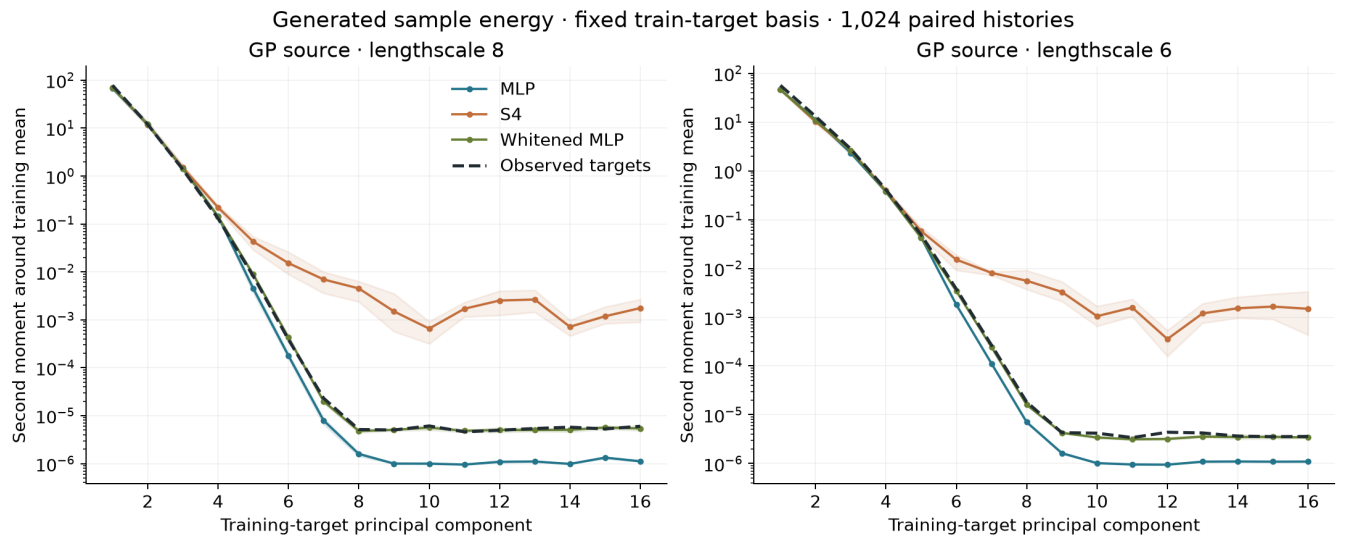
Original checkpoint head	CRPS ↓	Energy score ↓	90% coverage
MLP	0.4663	2.4132	91.1%
S4	0.4613	2.3920	90.3%
Whitened MLP	0.4606	2.3866	90.6%

These forecast scores evaluate the original trained models on new data. They are a separate experiment from the identity-history training control; the rows must not be treated as forecast results of the newly trained models.

The original results and declared success flags are preserved. Follow-up metrics address relative behavior, not a retrospective replacement of the assignment’s pass criterion.

Smoothness can hide directional error

A curve may have nearly correct average roughness while allocating the wrong amount of energy to weak principal components. We therefore projected archived generated paths and targets into a common PCA fitted on training targets. The displayed quantity is the second moment about the training mean, including any bias; it is not conditional variance.



Archived base runs at both GP lengthscales. All 16 target PCs are retained. The PC index orders variance and should not automatically be interpreted as Fourier frequency.

Lengthscale	Head	Generated	Target	Ratio
8	MLP	1.81e-06	7.14e-06	0.253
8	S4	0.00241	7.14e-06	338
8	WHITENED MLP	6.63e-06	7.14e-06	0.93
6	MLP	1.28e-05	3.17e-05	0.405
6	S4	0.00258	3.17e-05	81.4
6	WHITENED MLP	2.87e-05	3.17e-05	0.907

Mean second moments per component in PCs 7-16. Large ratios have a very small denominator: they are not overall forecast-quality ratios. MLP suppresses part of this tail, whereas the tested S4 overproduces it; whitening is closer to the observed tail.

Implementation recheck. All 49 earlier runs passed saved-array verification. Across 15 trained S4 layers, the independent complex reference differs from the real float64 kernel by at most 8.4e-16. For 15 complete GP heads, CPU/MPS maximum output disagreement is 6.7e-06. Device roundoff does not account for the observed tail discrepancy.

Separate the head from the loss weighting

Both heads receive the same complete 32-point normalized history through an identity encoder. The forecast horizon is 16; GP lengthscale is 8. MLP has 63,352 parameters and S4 has 63,169. S4 retains three temporal blocks with state size 16. This removes the separately learned history-encoder confound within the new suite.

Raw loss averages squared velocity errors. **Balanced loss** projects the error onto training velocity PCs and divides each squared component by its regularized training variance. Network inputs and outputs remain in temporal coordinates: S4 never convolves across PCA indices.

The floor is $\max(10^{-6} \times \text{leading variance}, 10^{-8})$; 10 of 16 velocity components are floored. The weighting is therefore regularized, not exact inverse variance in every direction.

Stage	Protocol
Data	Fresh GP draw: 32,768 train / 4,096 validation / 8,192 test; data seed 67041
Pilots	Each of four cells: 4,000 updates at LR 0.001 and 0.0003, seed 20; validation only
Selection	All four cells selected LR 0.001; selection locked before final test evaluation
Final fits	Seeds 10, 11, 12; 8,000 updates; batch 256; best validation checkpoint per run
Generation	1,024 paths per run; 64 Euler steps; matched histories and source draws

Head	Loss	PC1 MSE	Residual MSE	Roughness ratio
MLP	raw	14 ± 0.48	0.04669 ± 0.00095	1.01 ± 0.041
MLP	balanced	85.24 ± 0.59	0.04389 ± 0.0012	0.8659 ± 0.039
S4	raw	14.26 ± 0.051	0.05706 ± 0.0035	1.097 ± 0.076
S4	balanced	22.72 ± 1.6	0.06059 ± 0.00041	0.9471 ± 0.022

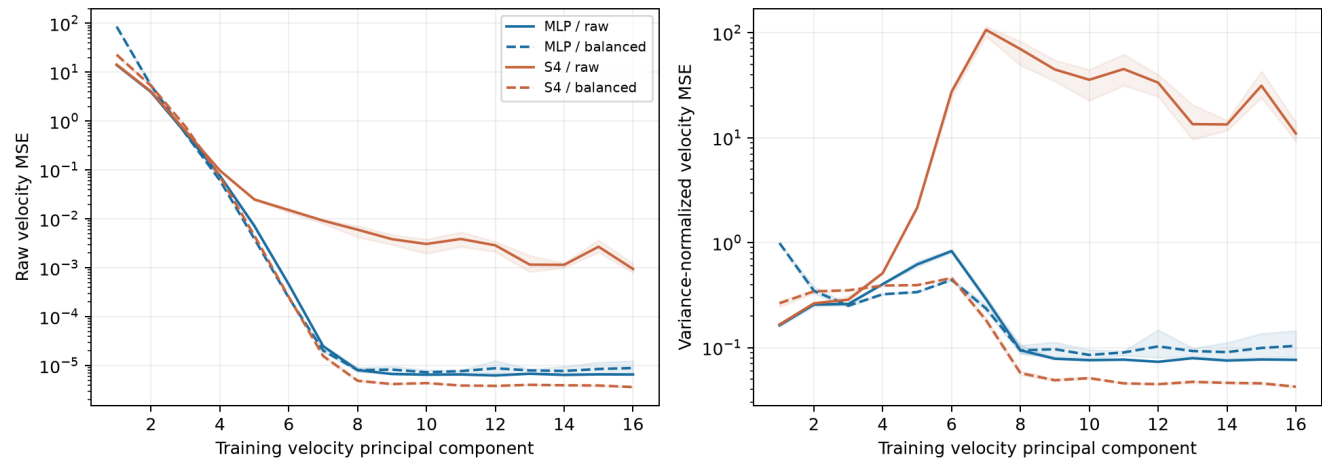
Paired contrast	Metric	Difference	95% interval
S4 minus MLP (raw)	Raw residual	+0.01037	[+0.007375, +0.01382]
S4 minus MLP (balanced)	Raw residual	+0.0167	[+0.01559, +0.01749]
S4 minus MLP (raw)	Normalized residual	+30.9	[+24.23, +36.9]
S4 minus MLP (balanced)	Normalized residual	-0.01692	[-0.02445, -0.01103]

Paired bootstrap over only three optimization seeds, conditional on this single GP dataset. Intervals are exploratory, with limited seed-level resolution; negative differences favor S4 for the stated error metric.

Different data, seeds, conditioning and budgets separate this suite from the original experiment. Their difference cannot identify the effect of removing the encoder alone. Two learning rates and a single pilot seed do not establish optimal tuning or convergence.

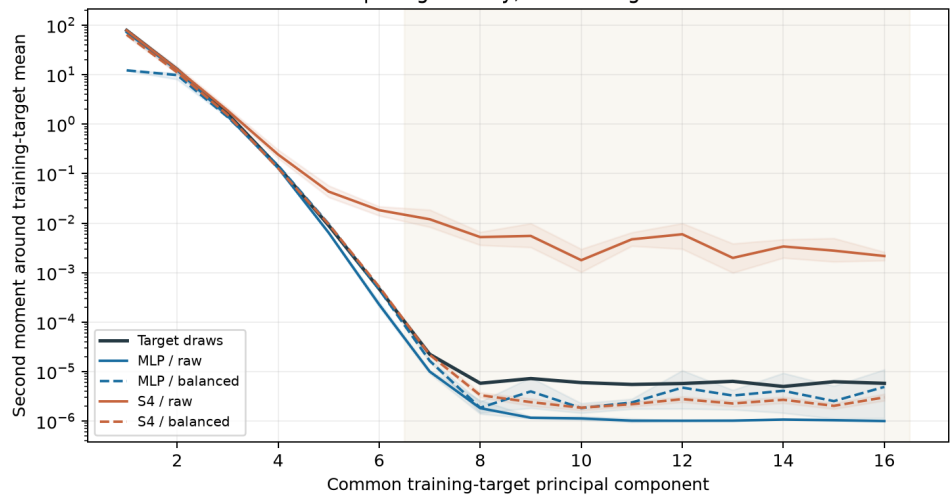
Does balancing also change generated paths?

Identity history: head × loss control (mean and seed range)



Velocity prediction: mean and range over three final seeds. Right: errors divided by regularized training velocity variances. A smaller normalized error need not imply a smaller raw error.

Generated-path geometry; shaded region is PCs7-16



Generated paths projected into a separate PCA fitted on training TARGETS. This diagnostic has no variance floor. Target and velocity PCA are distinct bases and are not interchangeable.

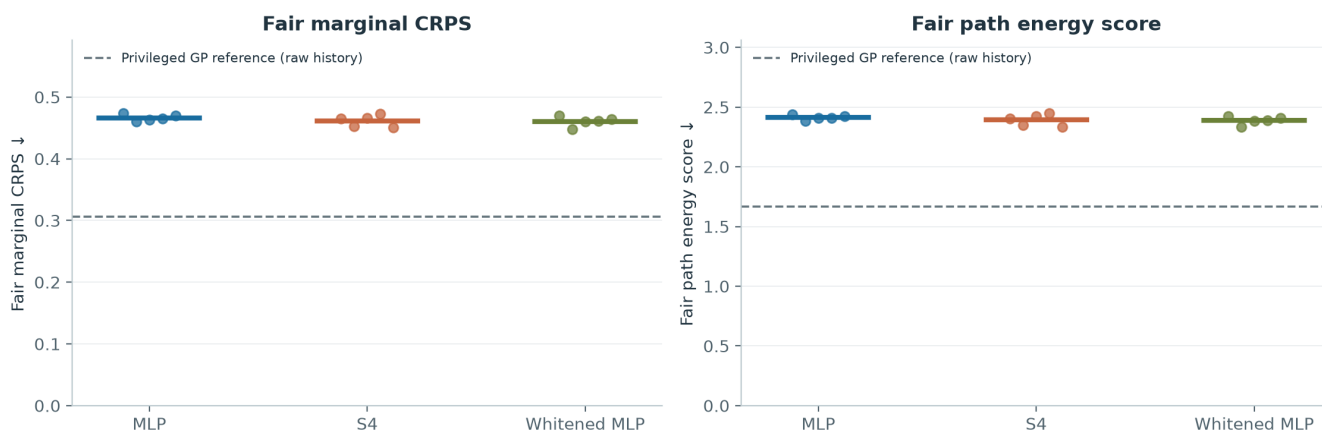
Head / objective	Normalized residual MSE	Generated / target tail moment
MLP / raw	0.2222	0.267
MLP / balanced	0.1746	0.6061
S4 / raw	31.12	597.2
S4 / balanced	0.1577	0.5851

Tail ratios sum second moments in target PCs 7-16 and average the ratio over three seeds. The target sum is only 7.6e-05. Values near one match this small-direction statistic, not full conditional calibration. All raw and normalized results are retained.

Evaluate the distribution, not only its texture

Each of the 15 original GP checkpoints predicts 64 futures for the same 256 fresh histories. Forecasting uses 128 Euler steps, with common source samples across heads and seeds. Fair marginal CRPS measures coordinatewise predictive quality; fair energy score measures the joint forecast path. Both use off-diagonal ensemble corrections and favor smaller values.

Conditional forecast scores on 256 fresh histories



Dots: five trained seeds; thick line: their mean. 64 samples per history, Euler-128. Lower is better.

Dots represent the five fixed training seeds. The exact GP reference conditions on raw history, information unavailable to these normalized-history networks; it is not an attainable bound for them.

Head	80% coverage	80% width	90% coverage	90% width
MLP	83.5%	2.115	91.1%	2.802
S4	81.6%	2.058	90.3%	2.761
Whitened MLP	82.4%	2.088	90.6%	2.781

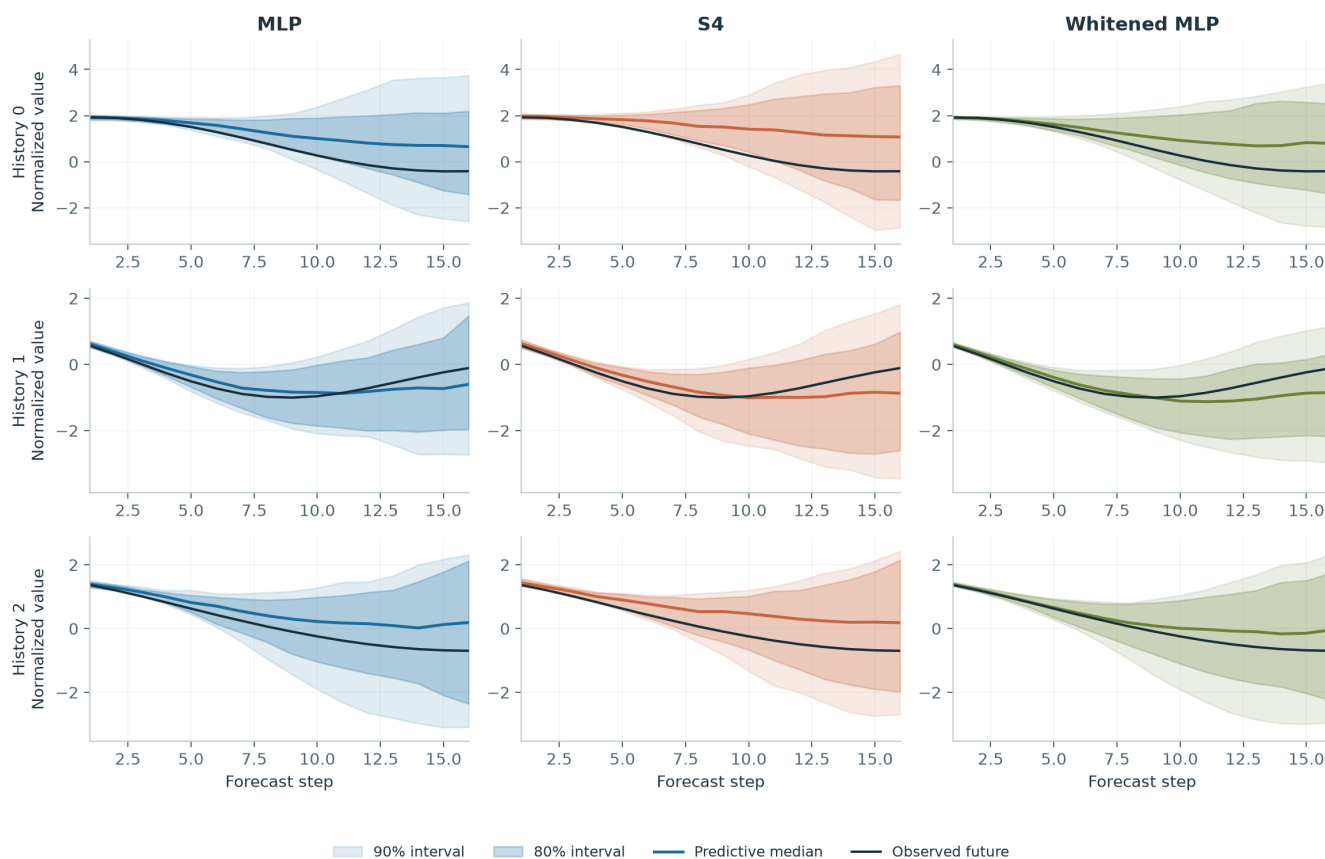
Coverage is marginal, averaged across histories and forecast coordinates. Width is in normalized units. Narrower intervals are only useful when considered alongside coverage and proper scores; they are not automatically better.

Contrast	Score	Difference	95% interval
S4 – MLP	crps	-0.0050	[-0.0168, +0.0064]
S4 – MLP	energy_score	-0.0212	[-0.0762, +0.0323]
WHITENED MLP – MLP	crps	-0.0057	[-0.0139, +0.0022]
WHITENED MLP – MLP	energy_score	-0.0266	[-0.0663, +0.0113]

Paired bootstrap resamples the 256 histories after averaging scores over the five fixed training seeds. These exploratory intervals condition on these models and source ensembles; they do not quantify new-training-data uncertainty and are not multiplicity-adjusted.

The same three histories for every head

First three fresh contexts: seed 0, fixed before evaluation



The first three fresh contexts were fixed before evaluation. Original seed 0; shared vertical scale across heads within a row. Bands are marginal 80% and 90% intervals, not simultaneous path bands. These examples illustrate behavior; aggregate scores use all 256 histories.

Does the Euler step count change the conclusion?

Head	Endpoint RMSE	Δ CRPS	Δ energy
MLP	0.01366	-5.582e-05	-0.0003616
S4	0.01273	-3.462e-07	+0.0001151
WHITENED MLP	0.0133	-0.0001548	-0.001042

Euler-256 minus Euler-128, paired on the same 64 source samples for the first 32 fresh histories, seed 0. This checks a numerical sensitivity on a fixed subset; it is not a new head-selection experiment. Endpoint RMSE compares the two integrations, not predictions against the observed future.

What the evidence supports

The strongest finding is an objective-dependent difference. With ordinary loss, normalized residual velocity MSE is 0.222 for MLP and 31.122 for S4. With balanced loss it is 0.175 and 0.158, respectively. All three paired seed differences have the corresponding sign. Raw residual error still favors MLP under both objectives.

Balancing has a cost. S4's generated/target tail second-moment ratio falls from 597 to 0.585, removing the excess but undershooting the target. Its mean PC1 error rises from 14.26 to 22.72; balanced MLP rises from 14.00 to 85.24. This supports studying the head and loss together, not declaring S4 uniformly superior. Ten of sixteen velocity directions use the variance floor; weighting is regularized.

Forecast quality remains unresolved between heads. For the original models on fresh histories, CRPS is 0.4663 (MLP), 0.4613 (S4) and 0.4606 (whitened MLP). All four declared proper-score comparisons against MLP have 95% paired-history intervals containing zero. This is insufficient evidence of superiority, not evidence of equivalence. Marginal 90% coverage is 90.3–91.1%; it does not establish conditional or simultaneous calibration. Euler-128/256 score changes on the fixed subset are small.

Recommendation. Retain the raw and whitened MLP baselines; keep balanced S4 as a useful comparison, not a proven replacement. A subsequent study should prospectively test intermediate directional weights and multiple independent training datasets, evaluating proper forecast scores alongside per-PC errors. Full coordinate whitening and loss-only balancing are different interventions.

Scope and assignment. All requested original tasks and outcomes are preserved. The two-sided raw 20% equality rule was our declared operational interpretation of an ambiguous instruction, not the only possible reading; follow-up findings do not redefine it. These small synthetic models, three new training seeds and 256 fresh evaluation histories cannot establish a universal architecture ranking. Independent audits verified saved arrays, checkpoint selection and provenance; all 34 tests passed, including CPU/MPS checks.

Reproduce and inspect

Artifact	Purpose
configs/controlled_followup.json	Fixed head-by-loss training protocol
configs/ensemble_evaluation.json	Fresh forecast evaluation and Euler sensitivity
scripts/check_followup.py	Independent saved-array and provenance checks
results/controlled_followup/	Pilots, selected rates, final checkpoints and samples
results/ensemble-evaluation/	Shared inputs, ensembles and per-history scores
output/controlled-followup/ output/ensemble-analysis/	Machine-readable results and figures

Assignment and provenance: [references/project-assignment.pdf](#); [references/project-plan.pdf](#); [docs/model_provenance.md](#); [docs/followup_independent_audit.md](#). The original task report and its success flags remain in [output/pdf/research-report.pdf](#). Implementation uses the pinned upstream references documented there.